

# Pallavi Reddy Rangasamudram


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## PROFESSIONAL SUMMARY


Systems Integration & Validation Engineer with hands-on experience across embedded systems, electronics, automation, system integration and validation. Experienced at the hardware–software interface, integrating controllers, sensors, communication networks and test systems from component level through system validation. Practical experience with embedded C/C++, CAN/CANoe, closed-loop control, HIL testing, robotic system integration, troubleshooting and automotive cybersecurity workflows. Structured, analytical and experienced in cross-functional engineering environments.


## TECHNICAL SKILLS

- **Systems Integration & Automation:** Hardware–software integration, robotic system integration, 6-DOF robotic platforms, system commissioning, calibration, motion verification, closed-loop control, automated test systems, BOM management
- **Embedded Systems & Electronics:** Embedded C/C++, ARM Cortex-M4, Real-time firmware, interrupt-driven systems, Sensor and actuator integration, Electronics debugging, Signal analysis
- **Testing & Validation:** HIL testing, Functional and limit testing, system validation, ECU state management, signal behaviour and timing verification, troubleshooting, root-cause analysis, Thermal validation (-60°C to +80°C)
- **Communication & Automotive:** CAN, LIN, K-line (exposure), CAN transceivers, ECU communication, CAN signal monitoring and debugging
- **Programming:** C/C++, Python (basic), MATLAB, CAPL
- **Engineering Tools:** Vector CANoe/CANalyzer, Codebeamer ALM, Autodesk Fusion 360, Oscilloscope, Logic Analyzer, Multimeter, Microsoft Office
- **Automotive Engineering & Standards:** ASPICE, ISO/SAE 21434, UNECE R155, automotive electronics, cybersecurity workflows and traceability
- **Documentation:** Technical documentation, commissioning procedures, installation guides, calibration records, validation reports, structured reporting
- **Languages:** English (C1 – fluent), German (A2 – actively improving), Telugu (native)

## PROFESSIONAL EXPERIENCE

**Minebea Access Solutions GmbH** Erdweg, Germany

***Master Thesis – Robotic Test Automation for Automotive Door-Handle ECU*** Oct 2025 – May 2026

- Led the mechanical assembly, installation and commissioning of a 6-DOF robotic test automation system, including component sourcing, BOM management, 3D-printed components and system-level integration.
- Performed robotic system commissioning including motor installation, gearbox assembly, encoder calibration, electrical integration and motion verification; investigated mechanical failures and implemented corrective actions to improve system reliability.
- Developed a closed-loop HIL test architecture integrating an Arduino GIGA R1-based controller with Vector CANoe for real-time evaluation of robotic actions against ECU signal feedback.
- Implemented CAN communication using a TJA1051 transceiver and interrupt-driven C/C++ firmware; investigated CAN signal integrity issues including termination and CANH/CANL behaviour using an oscilloscope.
- Designed and validated an end-effector probe for capacitive and NFC sensor interaction, evaluating different material configurations across force, temperature (-40°C to +80°C) and grounding conditions.

- Implemented ECU state monitoring in CANoe to analyse wake/sleep behaviour and initialization timing, supporting reliable validation of passive-entry and related body-domain functions.
- Worked closely with R&D, testing and mechanical engineering teams during system integration, commissioning and validation activities.
- Created technical documentation including commissioning procedures, installation guides, calibration records and validation reports to support engineering handover.

***Working Student – R&D Electronics Department*** Jan 2025 – Sep 2025

- Developed a Resource Planning Tracker in Codebeamer ALM, providing visibility into sprint progress, team availability and hardware development milestones across multiple project streams.
- Documented the tool architecture and development progress to support knowledge transfer and continuity for future development.

***Student Intern / Project Thesis – Automotive Cybersecurity*** Jul 2024 – Oct 2024

- Developed an automated CAL (Cybersecurity Assurance Level) inheritance framework across the ASPICE V-Model in Codebeamer ALM, incorporating TARA-based impact and attack-feasibility evaluation in accordance with ISO/SAE 21434 and UNECE R155.
- Automated traceability across system and software design layers, supporting structured cybersecurity validation and reducing manual effort and potential errors in the assessment workflow.

**Persistent Systems Ltd.** Pune & Hyderabad, India
***Associate Software Engineer / Intern*** Feb 2022 – Nov 2022

- Supported software development and structured testing activities within a cross-functional engineering environment.

## EDUCATION

**Ostbayerische Technische Hochschule (OTH) Regensburg** Regensburg, Germany
*Master of Engineering: Electrical and Microsystems* Oct 2023 – Expected Nov 2026
GPA: 2.4

**Sreyas Institute of Engineering and Technology** Hyderabad, India
*Bachelor of Technology: Electronics and Communication Engineering* Aug 2018 – Jul 2022
GPA: 2.1

## EXTRACURRICULAR & VOLUNTEERING

**CRICKET – Regensburg Women's Cricket, Germany:** Representing the Regensburg women's cricket team and volunteering in cricket promotion and training sessions at OTH Regensburg and local venues. Supporting event coordination and coaching, with participation in regional tournaments seasonally.

**CAMPUS RADIO – Sreyas Institute of Engineering and Technology, India:** Radio Jockey and member of the drafting team. Hosted a weekly radio programme, presented English news broadcasts and contributed to programme planning and content development.

**COMMUNITY SERVICE – India:** Organised and coordinated educational outreach activities for children, including nutrition awareness and initiatives aimed at improving access to learning through community engagement.